Curriculum Vitae

Ludovico T. Giorgini

C.L.E. Moore Instructor in Mathematics | Massachusetts Institute of Technology

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About Me

I am an applied mathematician developing data-driven reduced-order models for complex multiscale systems. My research combines score-based generative modeling, stochastic dynamics, response theory, and neural differential equations to construct computationally efficient models that preserve observed statistics, temporal correlations, and responses to perturbations, with applications in fluid dynamics, geophysics, and statistical physics.

Appointments

Massachusetts Institute of Technology — Cambridge, MA, USA
C.L.E. Moore Instructor in Mathematics — Sep 2024–present

Nordita — Stockholm, Sweden
Postdoctoral Researcher — Jun 2024–Sep 2024

Research Experience

Aeolus Labs — San Francisco, USA
Research Consultant — Jul 2025–Aug 2025

- Topic: Response-guided scenario analysis for tropical cyclones: adapting score-based statistical-response methods to the forecasting pipeline and prototyping low-latency ensemble predictors.

Massachusetts Institute of Technology — Cambridge, MA, USA
Research Intern — Jul 2023

- Topic: Construction of response functions using generative modeling
- Advisors: Dr. Andre Souza, Prof. Raffaele Ferrari

Woods Hole Oceanographic Institution — Falmouth, USA
GFD Fellow — Jun–Aug 2022

- Topic: Statistical learning of multidimensional dynamical systems
- Advisors: Prof. Peter Schmid (KAUST), Dr. Andre Souza (MIT)

Atomki, CEA, Missouri S&T — Debrecen, Hungary; Saclay, France; Rolla, USA
Research Intern — Nov 2018–Feb 2019 / Sep 2022 / Jul 2023 / Aug 2024

- Topic: Calculation of the critical exponents of a ϕ^4 theory
- Advisors: Prof. U.D. Jentschura (Missouri S&T), Prof. J. Zinn-Justin (CEA)

Education

Nordita and Stockholm University — Stockholm, Sweden
PhD, Theoretical Physics — Mar 2019–May 2024

- Thesis: A Serendipitous Journey through Stochastic Processes
- Advisor: Prof. John Wettlaufer (Yale University and Nordita)

Sapienza University of Rome — Rome, Italy
M.Sc., Theoretical Physics — Oct 2016–Oct 2018

- Thesis: High-Order Behavior of the Correlation Functions of a ϕ^4 theory
- Advisor: Prof. Giorgio Parisi
- Graduation grade: 110/110 cum laude

Sapienza University of Rome — Rome, Italy
B.Sc., Physics — Oct 2013–Oct 2016

- Thesis: The Birth of the Universe: from Friedmann's Equations to the Inflationary Model
- Advisor: Prof. Alessandro Melchiorri
- Graduation grade: 110/110

Teaching Experience

Massachusetts Institute of Technology Cambridge, MA, USA
Primary Lecturer Fall 2026

- 18.305 Advanced Analytic Methods in Science and Engineering — Fall 2026

Massachusetts Institute of Technology Cambridge, MA, USA
Primary Lecturer Spring 2026

- 18.354J Nonlinear Dynamics II: Continuum Systems (graduate) — Spring 2026

Massachusetts Institute of Technology Cambridge, MA, USA
Teaching Assistant Sep 2024–May 2026

- Multivariate Calculus (undergraduate, first cycle) — Sep 2025–May 2026
- Differential Equations (undergraduate, first cycle) — Jan 2025–May 2025
- Differential Equations (undergraduate, first cycle) — Sep 2024–Dec 2024

Stockholm University Stockholm, Sweden
Teaching Assistant Jan 2020–Mar 2023

- Statistical Physics (graduate, second cycle) — Jan 2023–Mar 2023
- Mathematical Methods for Physicists (graduate, second cycle) — Sep 2022–Nov 2022
- Analytical Mechanics (graduate, second cycle) — Jan 2022–Mar 2022
- Mathematical Methods for Physicists (graduate, second cycle) — Sep 2021–Nov 2021
- Mathematical Methods for Physicists (graduate, second cycle) — Sep 2020–Nov 2020

Stockholm University Stockholm, Sweden
Higher-education teaching training course (Faculty of Science) Autumn 2020

- Provider: Department of Mathematics & Science Education and the Centre for the Advancement of University Teaching (CeUL).
- Workload: ~50 hours of higher-education pedagogy; structured meetings plus a CeUL pedagogical workshop.
- Objective: develop skills to plan, deliver, and evaluate teaching through practice-based tasks (lesson planning, peer feedback, classroom observation, reflective essay).

Supervision and Mentoring

Massachusetts Institute of Technology Cambridge, MA, USA
Undergraduate Research Opportunities Program (UROP) Summer 2026

- Student: Tristan Sprole (MIT)
- Topic: Climate detection-and-attribution and response-operator methods

Massachusetts Institute of Technology Cambridge, MA, USA
Undergraduate Research Opportunities Program (UROP) Summer 2026

- Student: Laura Lerebours (MIT)
- Topic: Climate detection-and-attribution and response-operator methods

Massachusetts Institute of Technology Cambridge, MA, USA
Master's Thesis Project Apr 2025–Oct 2025

- Student: Giulio Del Felice (ETH Zurich)
- Topic: Integrating Score-Based Generative Modeling and Neural ODEs for Accurate Representation of Multiscale Chaotic Dynamics

Nordita Stockholm, Sweden
Research Internship Project Nov 2021–May 2022

- Student: Nash Keyes (Yale University)
- Topic: Stochastic Paleoclimatology: Modeling the EPICA Ice Core Climate Records

Nordita Stockholm, Sweden
Undergraduate Thesis Project Nov 2020–May 2021

- Student: Nash Keyes (Yale University)
- Topic: Stochastic Paleoclimatology: Modeling the EPICA Ice Core Climate Records

Nordita Stockholm, Sweden
Undergraduate Thesis Project Mar 2020–Jun 2021

- Student: Olle Bergström Jonsson (Stockholm University)
- Topic: Predicting rare events in multistable systems

Open-Source Software

- **DiscreteTransformers.jl** — GitHub. Julia package for time-series forecasting with transformers on discretized state spaces; clusters continuous signals into symbols and learns transition dynamics for ensemble forecasting and scenario generation.
- **ContinuousTransformers.jl** — GitHub. Transformers operating directly on continuous, delay-embedded sequences for next-step prediction, probabilistic ensembles, and compact representations.
- **FastSDE.jl** — GitHub. Speed-first Julia library for simulating stochastic/deterministic dynamical systems (additive/diagonal/correlated noise, StaticArrays fast path, multithreading).
- **ScoreEstimation.jl** — GitHub. KGMM-based score estimation: bisecting k-means + Gaussian mixture modeling with neural interpolation to build robust local score fields in high-dimensional dynamics.
- **ParameterCalibration.jl** — GitHub. GFDT-based parameter calibration toolkit: computes Jacobians of statistical observables from a single baseline run; supports drift/diffusion parameters with worked examples.
- **ScoreUNet1D.jl** — GitHub. Julia package that trains a 1D U-Net via denoising score matching and validates its learned score by running Langevin SDE on the Kuramoto–Sivashinsky system.
- **StateDependentMobility.jl** — GitHub. Julia package for learning and validating state-dependent mobility tensors from finite-lag trajectories and conditional scores.

Invited Talks and Conference Presentations

- **2026** — SIAM Conference on Uncertainty Quantification, Minneapolis, MN, USA. Contributed talk: "Response Theory via Score Modeling" (Mar 22).
- **2026** — Department of Mathematics, University of Rochester, Rochester, NY. Invited talk: "Learning Dynamics from Statistics: A Score-Based Approach for Modeling Complex Systems" (Feb 9).
- **2025** — Statistical Mechanics of the Climate System and of Ecosystems, Leicester, UK. Invited lecture: "Response Theory via Score Modeling" (Jul 1–3).
- **2025** — University of Rome "Tor Vergata", Rome, Italy. Invited seminar: "Response Theory via Score Modeling" (Jun 24).
- **2025** — APS Division of Fluid Dynamics (78th Annual Meeting), Houston, USA. Contributed talk: "From Memory-Driven Chaos to Markovian Dynamics in One-Dimensional Pilot-Wave Hydrodynamics" (Nov 24).
- **2024** — MIT Physical Mathematics Seminar, Cambridge, MA, USA. Invited seminar: "Response Theory via Generative Score Modeling" (Sep 24).
- **2024** — AGU Fall Meeting, Washington, DC, USA. Poster: "Response Theory via Generative Score Modeling" (Dec 12).
- **2024** — SIAM UQ24, Trieste, Italy. Contributed talk: "Tackling the Curse of Dimensionality: A Generative Modeling Approach to Linear Response Theory" (Feb 27–Mar 1).
- **2023** — AGU Fall Meeting, San Francisco, USA. Poster: "Non-Gaussian Stochastic Dynamical Model for the El Niño Southern Oscillation" (Dec 14).
- **2023** — APS March Meeting, Las Vegas, USA. Contributed focus-session talk: "Non-Gaussian Stochastic Dynamical Model for the El Niño Southern Oscillation" (Mar 6).
- **2022** — AGU Fall Meeting, Chicago, USA. Poster: "Non-Gaussian Stochastic Dynamical Model for the El Niño Southern Oscillation" (Dec 14).
- **2021** — AGU Fall Meeting (hybrid), New Orleans, USA. Poster: "Modeling the El Niño Southern Oscillation with Neural Differential Equations" (Dec 14).
- **2021** — ICML 2021 Time Series Workshop, Online. Contributed paper and oral presentation: "Modeling the El Niño Southern Oscillation with Neural Differential Equations" (Jul).
- **2020** — AGU Fall Meeting (virtual), Online. Poster: "A Stochastic ENSO Model for State Estimation, Uncertainty Quantification and Prediction" (Dec 14).
- **2019** — AGU Fall Meeting, San Francisco, USA. Poster: "Predicting Rare Events in Stochastic Resonance" (Dec 9).

Scholarships and Honors

- John Söderbergs scholarship (6989 SEK) — 2023–2024
- Elisabeth och Herman Rhodins scholarship (8017 SEK) — 2023–2024
- L. Namowitskys scholarship (13133 SEK) — 2022–2023
- Kobbs scholarship (11867 SEK) — 2022–2023
- GFD Fellowship (9494 USD) — Summer 2022
- Computational grant from SNIC at HPC2N (60000 core hours) — 2021–2022
- Smitts scholarship (27333 SEK) — 2020–2021
- C.F. Liljevalch scholarship (7000 SEK) — 2020–2021

- Path of Excellence Program — Research project in Number Theory, Mathematics Department, Sapienza University of Rome, supervised by Prof. Valentina Barucci — 2015–2016

Funding Proposals

- *Response-Operator Methods for Climate Detection and Attribution* (MIT Global Seed Funds; budget request: 30,000 USD).
- *Integrating the Generalized Fluctuation–Dissipation Theorem and Generative Score Modeling* (NSF program 24-569, "Mathematical Foundations of Artificial Intelligence"; budget request: 1.2M USD).
- *Score-Based Reduced-Order Modeling of Arctic Sea Ice* (NSF program PD 16-1266, "Applied Mathematics"; budget request: 1.2M USD).

Academic Service

- Referee for Physical Review Letters, Physical Review E, Physica D, IEEE Transactions on Information Theory, Philosophical Transactions A, and Journal of Chemical Physics.
- PhD representative at Nordita — Nov 2020–Dec 2023

Skills

Programming Languages
Python, Julia, MATLAB, C/C++, Mathematica, LaTeX

Languages
Italian (native), English (C2), German (B2)

Personal Interests

Transverse flute and cello playing, salsa, bachata, and tango dancing, sommelier.

Publications

Publications Related to Current Research

1. **L. T. Giorgini**, "Score-Based Modeling of Effective Langevin Dynamics," *Physical Review E* **114**, L012102 (2026).
2. **L. T. Giorgini**, "Conditional Score-Based Modeling of Effective Langevin Dynamics," arXiv:2604.23952 (2026).
3. **L. T. Giorgini**, T. Bischoff, and A. N. Souza, "KGMM: A K-means Clustering Approach to Gaussian Mixture Modeling for Score Function Estimation," *Physica D: Nonlinear Phenomena* **495**, 135274 (2026).
4. G. Del Felice and **L. T. Giorgini**, "Integrating Score-Based Generative Modeling and Neural ODEs for Accurate Representation of Multiscale Chaotic Dynamics," *Chaos* **36**, 063143 (2026).
5. **L. T. Giorgini**, F. Falasca, and A. N. Souza, "Predicting Forced Responses of Probability Distributions via the Fluctuation–Dissipation Theorem and Generative Modeling," *Proceedings of the National Academy of Sciences* **122**, e2509578122 (2025).
6. **L. T. Giorgini**, T. Bischoff, and A. N. Souza, "Statistical Parameter Calibration via the Generalized Fluctuation Dissipation Theorem and Generative Modeling," arXiv:2509.19660 (2025).
7. **L. T. Giorgini**, T. Bischoff, and A. N. Souza, "Reduced-Order Modeling of Cyclo-Stationary Time Series Using Score-Based Generative Methods," arXiv:2508.19448 (2025).
8. **L. T. Giorgini**, K. Deck, T. Bischoff, and A. N. Souza, "Response Theory via Generative Score Modeling," *Physical Review Letters* **133**, 267302 (2024).

Other Publications

9. P. Patra, **L. T. Giorgini**, and J. S. Wettlaufer, "Stochastic Coupling of Climate Variables and Ice Volume over the Late Pleistocene Glacial Cycles," arXiv:2603.26937 (2026).
10. **L. T. Giorgini**, A. N. Souza, D. Lippolis, P. Cvitanović, and P. J. Schmid, "Learning Dissipation and Instability Fields from Chaotic Dynamics," *Physica D: Nonlinear Phenomena* **481**, 134865 (2025).
11. **L. T. Giorgini**, A. N. Souza, and P. J. Schmid, "Reduced Markovian Models of Dynamical Systems," *Physica D: Nonlinear Phenomena* **470**, 134393 (2024).
12. **L. T. Giorgini**, W. Moon, and J. S. Wettlaufer, "Analytical Survival Analysis of the Non-autonomous Ornstein–Uhlenbeck Process," *Journal of Statistical Physics* **191**, 138 (2024).

13. U. D. Jentschura and **L. T. Giorgini**, “Enhanced and Generalized One-Step Neville Algorithm: Fractional Powers and Access to the Convergence Rate,” *Computer Physics Communications* **303**, 109280 (2024).

14. **L. T. Giorgini**, U. D. Jentschura, E. M. Malatesta, T. Rizzo, and J. Zinn-Justin, “Instantons in ϕ^4 Theories: Transseries, Virial Theorems, and Numerical Aspects,” *Physical Review D* **110**, 036003 (2024) (Editors’ Suggestion).

15. N. D. B. Keyes, **L. T. Giorgini**, and J. S. Wettlaufer, “Stochastic Paleoclimatology: Modeling the EPICA Ice Core Climate Records,” *Chaos* **33**, 093132 (2023).

16. **L. T. Giorgini**, R. Eichhorn, M. Das, W. Moon, and J. S. Wettlaufer, “Thermodynamic Cost of Erasing Information in Finite Time,” *Physical Review Research* **5**, 023084 (2023).

17. **L. T. Giorgini**, W. Moon, N. Chen, and J. S. Wettlaufer, “Non-Gaussian Stochastic Dynamical Model for the El Niño Southern Oscillation,” *Physical Review Research* **4**, L022065 (2022).

18. **L. T. Giorgini**, U. D. Jentschura, E. M. Malatesta, G. Parisi, T. Rizzo, and J. Zinn-Justin, “Correlation Functions of the Anharmonic Oscillator: Numerical Verification of Two-Loop Corrections to the Large-Order Behavior,” *Physical Review D* **105**, 105012 (2022).

19. W. Moon, **L. T. Giorgini**, and J. S. Wettlaufer, “Analytical Solution of Stochastic Resonance in the Nonadiabatic Regime,” *Physical Review E* **104**, 044130 (2021).

20. **L. T. Giorgini**, S. H. Lim, W. Moon, N. Chen, and J. S. Wettlaufer, “Modeling the El Niño Southern Oscillation with Neural Differential Equations,” Time Series Workshop at ICML 2021 (2021).

21. S. H. Lim, **L. T. Giorgini**, W. Moon, and J. S. Wettlaufer, “Predicting Critical Transitions in Multiscale Dynamical Systems Using Reservoir Computing,” *Chaos* **30**, 123126 (2020).

22. **L. T. Giorgini**, W. Moon, and J. S. Wettlaufer, “Analytical Survival Analysis of the Ornstein–Uhlenbeck Process,” *Journal of Statistical Physics* **181**, 2404–2414 (2020).

23. **L. T. Giorgini**, U. D. Jentschura, E. M. Malatesta, G. Parisi, T. Rizzo, and J. Zinn-Justin, “Two-Loop Corrections to the Large-Order Behavior of Correlation Functions in the One-Dimensional N-Vector Model,” *Physical Review D* **101**, 125001 (2020).

24. **L. T. Giorgini**, S. H. Lim, W. Moon, and J. S. Wettlaufer, “Precursors to Rare Events in Stochastic Resonance,” *EPL* **129**, 40003 (2020) (Editor’s Choice).